David X. Lin

davidlin@cs.utexas.edu ◇ davidxlin.github.io

Education

Ph.D., The University of Texas at Austin, Austin, TX — August 2026 – present
Computer Science

M.S., Cornell University, Ithaca, NY — August 2024 – August 2026
Computer Science
Thesis: *Robust Equilibria in Shared Resource Allocation*
Thesis Committee: Siddhartha Banerjee (co-chair), Éva Tardos (co-chair), David Shmoys

B.A., Cornell University, Ithaca, NY — August 2019 – May 2023
Mathematics, Magna Cum Laude
Computer Science, Summa Cum Laude

Publications

- **David X. Lin**, Giannis Fikioris, Siddhartha Banerjee, and Éva Tardos. Robust Resource Allocation via Competitive Subsidies. In *Proceedings of the Innovations in Theoretical Computer Science Conference (ITCS)*, 2026.
- **David X. Lin**, Giannis Fikioris, Siddhartha Banerjee, and Éva Tardos. Robust Equilibria in Shared Resource Allocation via Strengthening Border's Theorem. In *Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2026.
- **David X. Lin**, Daniel Hall, Giannis Fikioris, Siddhartha Banerjee, and Éva Tardos. Online Resource Sharing: Better Robust Guarantees via Randomized Strategies. In *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 179–186, 2025.
- Hyunchul Lim, **David Lin**, Jessica Tweneboah, and Cheng Zhang. HandyTrak: Recognizing the Holding Hand on a Commodity Smartphone from Body Silhouette Images. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 1210–1220, 2021.

Preprints

- **David X. Lin**, Chido Onyeze, Giannis Fikioris, Siddhartha Banerjee, and Éva Tardos. Non-Monetary Mechanisms for Stochastic Scheduling via Sampled Pairwise Externality Compensation. arXiv:2610.25067, 2026.
- Chido Onyeze, **David X. Lin**, Siddhartha Banerjee, and Éva Tardos. Dynamic Allocation of Public Goods with Approximate Core Equilibria. arXiv:2511.04817, 2025.

Teaching Experience

Teaching Assistant at Cornell University

- CS 4850: Probability, Vectors, and Matrices in Computing — Spring 2026
- CS 4820: Introduction to Analysis of Algorithms — Fall 2021, Fall 2022, Fall 2025
- CS 2800: Mathematical Foundations of Computing — Spring 2021, Spring 2022, Spring 2025
- CS 4830: Introduction to Cryptography — Fall 2024
- CS 2802: Discrete Structures - Honors — Fall 2020, Spring 2023

Industry Experience

Software Engineer, Google — July 2023 – August 2024
- Built system software for Android XR.

Software Engineering Intern, Google — Summer 2022
- Built user-facing augmented reality experiences.

Software Development Engineering Intern, Amazon — Summer 2021
- Built tools to trace Amazon API Gateway requests.

Awards and Honors

- Outstanding Teaching Assistant Recognition — Spring 2025
- Top 500 scorer in the William Lowell Putnam Mathematics Competition — 2022
- Phi Beta Kappa Honor Society, Theta Chapter of New York at Cornell University — 2022